

CG TECHNICAL DIRECTOR

The Technical Director (TD) Training Program is designed to develop highly skilled TDs capable of handling the technical challenges in film and animation production. Participants will gain hands-on experience in pipeline development, rigging, rendering, and tools development, using industry-standard software like Maya, Houdini, and proprietary systems. The program emphasizes problem-solving, automation, and optimization within a collaborative production environment. Graduates will be prepared to manage and troubleshoot technical workflows, contributing effectively to high-end production pipelines.

JOB PROFILES

Pipeline Technical Director (Pipeline TD) · Character TD Artist · Rendering TD Artist · FX and Dynamics Artist · CG Tools Developer · FX Technical Director



TOTAL HOURS

1,404^{hrs}

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Minimum grade required



Where graduates work

Mercury Visual Solutions

AROMA Studios

Trend VFX

The Crew

Nutshell

Squids

Paradox VFX Studio

Dots House

Gimini Studios

Monkeys



Who may apply

Applications are open to graduates and final-year students from all academic disciplines at recognized universities, provided they complete their degree before the program begins. While applicants with backgrounds in Computer Science, Software Engineering, Computer Engineering, Communications Engineering, Mechatronics, Information Systems, and Computer Graphics are preferred, candidates from all other disciplines are warmly encouraged to apply. In line with our inclusive admissions policy, applicants with a minimum graduation grade of Fair are eligible for consideration.



Selection process

01 IQ & English Exam



Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam



Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview



Interview with the technical committee.

04 Interpersonal Interview



Candidates who pass the technical interview are invited to this final stage.

Program content – 1,404 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Fundamental Courses	171 hrs	12%	Track Orientation, CG Introduction, Cinematography: Color Theory and Light, Cinematography: Composition Theory, 3D Animation Essentials, Video Editing, Sound Editing and Mixing, Digital Arts Ai Creation Fundamentals, Film Language and Directing, Freelancing For Digital Arts, Script Writing
● Core Courses	876 hrs	62%	Agile Software Development Methodologies, AWS fundamentals for cg, Basics of rigging and character animation, CG Compositing Fundamentals For TD, Data structure and algorithms, Houdini fundamentals for TD, Introduction to AI and Machine Learning for digital arts, Introduction to houdini, Introduction to Maya, Introduction to Programming with C and C++, Introduction to Python Programming for CG Development, Introduction to Unreal, Look Development Essentials using Houdini, Look Development Essentials using Maya, Mathematics 1 (Mathematic for CG), Mathematics 2 : Probability and Statistics, Maya fundamentals for TD, Networking Fundamentals for CG, Procedural Modeling for TD, Python Applications, Research and Development in Computer Graphics, TD Portfolio Making, Technical Writing and Documentation, Unreal Programming Applications, Version Control Systems with Git, CG Web Fundamentals, Operating Systems
● Soft Skills	57 hrs	4%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor
● Graduation Project	300 hrs	21%	

Student Deliverable

Deliverables: To successfully complete the track, every student is required to submit a final production project that demonstrates professional-quality technical direction, pipeline development, automation, rigging, tools programming, and production workflow, meeting industry standards and preparing them for careers in film, animation, visual effects, game production, and freelance opportunities.

Software/Vendor Certificates (Upon Request): In addition to the final project, students may be required to obtain and submit official software certifications (or other designated vendor certifications) for specified courses, only if requested by the track leader during the program.

Before you apply

Basic knowledge of programming, computer graphics, and Python. Familiarity with Maya, Houdini, or Blender is recommended.

Sidefx Houdini

<https://www.sidefx.com/learn/>

Python

<https://www.youtube.com/watch?v=rfscVS0vtbw>

Maya

https://www.youtube.com/channel/UCHmAXsicpLK2EHMZo5_BtDA

Linear Algebra

https://www.khanacademy.org/math/linear-algebra?utm_

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.